Contact vs. Non-Contact Lighting: Do We Understand What Remote-PPG is Measuring?

Hongjia Liu[1], Huailei Lai[1], Huaijing Shu[1], Wenjin Wang[1,*]

***Abstract*— As photoplethysmography (PPG) technology evolves from traditional contact-based sensors to non-contact monitoring (remote-PPG), the fundamental mechanism of light-tissue interaction undergoes a tectonic shift. However, how these distinct illumination conditions influence the spatial formation of physiological signals remains insufficiently understood. This study presents a spatial analysis of remote-PPG signal metrics, including AC, DC, AC/DC, and multi-wavelength pulse transit time (MW-PTT), measured from fingers illuminated by contact and non-contact light sources. Spatial heatmaps are used to visualize pixel-level distributions of remote-PPG signal metrics for comparing the contact and non-contact illuminations in the setup, especially for understanding the tissue-light interaction difference in between. Results from 15 healthy subjects show that non-contact illumination produces uniform spatial distribution on their measured fingers, whereas contact illumination exhibits strong distance- and wavelength-dependent patterns. In the contact mode, AC/DC and MW-PTT demonstrate characteristic rise-and-fall trends in heatmaps, with peak locations shifting systematically across wavelengths, validating the wavelength-dependent optical penetration depth. These findings highlight the significant role of illumination configuration in remote-PPG: while the transition from contact to non-contact illumination introduces challenges, it also provides key benefits that can enhance circulatory monitoring (e.g. MW-PTT for microcirculation imaging).**

***Index Terms*— Remote photoplethysmography, multi-wavelength pulse transit time, illumination source, spatial heatmap analysis**

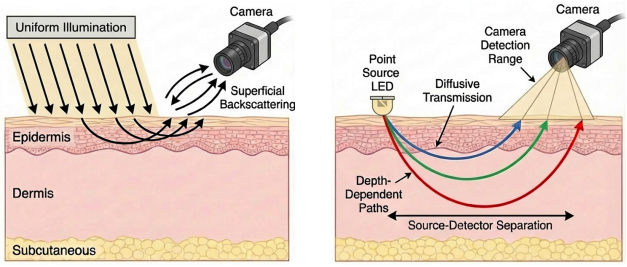
I. INTRODUCTION

Photoplethysmography (PPG) has long served as a fundamental optical technique for non-invasive physiological monitoring [1]. In recent years, advances in computer vision have boosted the evolution from contact-based monitoring to non-contact implementations, also called remote photoplethysmography (rPPG) [2]. Beyond basic heart rate tracking, multi-wavelength rPPG facilitates advanced physiological assessments, including oxygen saturation (SpO_2) [3], [4] and blood pressure estimation [5], [6]. The accuracy of these applications relies on specific signal metrics, including the pulsatile component (AC), the non-pulsatile component (DC), the normalized perfusion index (AC/DC), and the inter-wavelength phase difference known as Multi-Wavelength Pulse Transit Time (MW-PTT) [7].

This work is supported by the National Key R&D Program of China (2025YFE0123200, 2024YFC2417600), Shenzhen Key Industrial R&D Program (ZDCYKCX20250901092803004), Shenzhen Medical Research Fund (D2402011), Shenzhen Science and Technology Program (JSGGKQTD20221103174704003), and Guangdong ST program (2023ZT10Z002).

[1]Department of Biomedical Engineering, Southern University of Science and Technology, China.

[*]Corresponding author: Wenjin Wang, email: wangwj3@sustech.edu.cn



(a) Non-Contact Illumination (b) Contact Illumination

Fig. 1: Schematic of light-tissue interaction. (a) Non-contact mode showing uniform superficial backscattering. (b) Contact mode exhibiting depth-dependent diffusive photon transmission (e.g. banana-shape optical path).

As illustrated in Fig. 1, the transition from contact to non-contact physiological monitoring fundamentally alters the mechanism of light-tissue interaction. Described in [8], multiple scattering and absorption events govern light propagation in biological tissue. In the non-contact modes, ambient light is often assumed to be generally uniform, and the collected signal is dominated by backscattering from superficial skin layers. In contrast, contact-based PPG typically employs a point light source, transforming the tissue into a diffusive transmission medium, where photon migration relies on multiple scattering events to reach distal regions [9]. Despite these distinct optical behaviors, the majority of existing rPPG research continues to prioritize the development of backend signal processing algorithms, ranging from prior-knowledge-driven multi-wavelength combination [10], [11] to deep learning frameworks [12]. However, algorithmic improvement cannot fully compensate for physical limitations introduced at the frontend illumination geometry. While some recent studies enhances the quality of rPPG signal by adjusting the exposure time of camera [13], there remains a lack of systematic understanding on how the illumination condition, particularly the difference between contact and non-contact modes, shapes the spatial imaging characteristics of AC, DC, AC/DC and MW-PTT.

Theoretical simulations using Monte Carlo methods have suggested that source-detector separation significantly influences the probing depth in contact modes [14]. However, these theoretical models have rarely been validated against high-resolution spatial imaging data that compares both contact and non-contact illumination in reality directly. We hypothesize that these distinct optical mechanisms induce

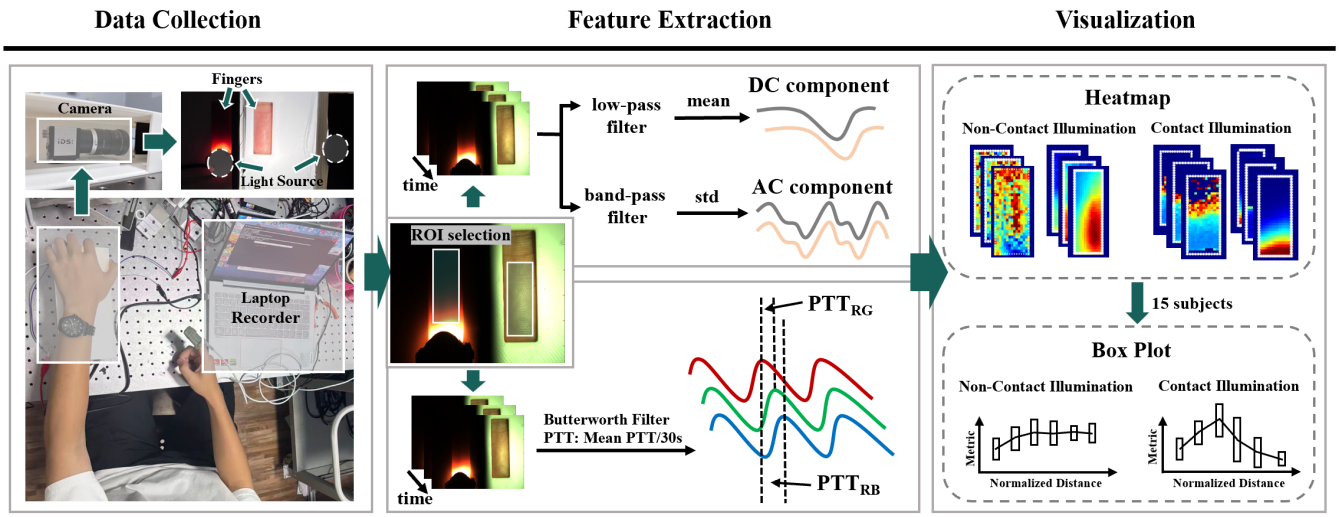


Fig. 2: The overall workflow of this research encompasses several stages, including video data acquisition, extraction of rPPG features, and visualization for comparing two illumination modes (contact versus non-contact light sources) using heatmaps and boxplots.

characteristic spatial variations in signal metrics (AC, DC, AC/DC and MW-PTT), which are critical for understanding signal formation and ensuring accurate physiological measurement.

To validate this hypothesis, we developed a single-camera imaging system capable of simultaneously operating under contact and non-contact illumination modes. By capturing video data from the same subject (i.e. the index and middle fingers) under both illumination conditions concurrently, we eliminated sensory and physiological variability to perform a fair comparison. Unlike traditional time-series analysis that focuses on vital signs measurement, we utilized spatial heatmaps to visualize and quantify the pixel-level spatial distributions of AC, DC, AC/DC, and MW-PTT, advancing our understanding of how perfusion imaging in rPPG is changed from contact to non-contact illumination.

II. MATERIALS AND METHODS

A. Experimental Setup and Protocol

A schematic diagram of the rPPG imaging setup is shown in Fig. 2. The data acquisition system was centered on a UI-3160CP-C-HQ industrial CMOS camera with a resolution of 960×600 pixels. Illumination was provided by 3 W full-spectrum 3535 white LEDs, each equipped with a 45° optical lens to generate a focused field. To enable a simultaneous comparison between contact and non-contact illumination conditions, the middle and index fingers of the subject's left hand were imaged concurrently at a distance of approximately 10 cm from the camera. The fingertip of the middle finger was placed in direct contact with the LED lens surface (contact mode) located at the bottom of the Region of Interest (ROI) mask. The index finger was positioned 5 cm above an identical light source (non-contact mode). In this configuration, the light source was positioned at the bottom-right corner relative to the finger. Raw video sequences were recorded at 100 FPS for rPPG signal analysis (AC, DC, AC/DC), and subsequently resampled to 200 FPS to provide sufficient temporal resolution for MW-PTT calculation.

A total of 15 healthy volunteers participated in this pilot (2 females, 13 males). The study was approved by the Institutional Review Board (IRB) of Southern University of Science and Technology (IRB no. 20230184), and written informed consent was obtained from all participants. The experiment was conducted in a dark chamber with only the designated LED light source activated to draw deterministic conclusions. During each 60-second trial, subjects were instructed to remain in a stationary state to minimize motion artifacts.

B. Methods

1) ROI Selection and Signal Extraction: As shown in Fig. 2, a ROI mask covering the target finger area was manually defined and remained identical for both contact and non-contact modes. The ROI was spatially segmented into 5×5 pixel grid blocks. The mean intensities of the Red, Green, and Blue channels within each block were calculated to generate the raw signal traces. Subsequently, spatial heatmaps were generated based on the amplitudes of the computed metrics of each trace.

2) Extraction of AC and DC components:

- DC component: A fourth-order Butterworth low-pass filter with a cutoff frequency of 0.6 Hz was applied to remove high-frequency noise. The filtered signal was averaged to obtain the DC component for the corresponding pixel blocks.
- AC component: A fourth-order Butterworth band-pass filter with a passband of [0.6, 3] Hz was utilized to isolate the raw AC signal, which was then normalized by dividing it by the corresponding DC component. The standard deviation (STD) of the raw AC signal was computed as AC, while the STD of the DC-normalized signal was defined as AC/DC.

3) Calculation of MW-PTT: The fundamental principle of MW-PTT lies in the wavelength-dependent optical penetration depth of skin tissue [7]. Longer wavelengths (e.g., Red) penetrate deeper into the dermis, interacting with deeper arterioles, whereas shorter wavelengths (e.g., Green and Blue) are confined to superficial capillaries [15]. By employing a high-speed camera, the temporal difference between the arrival of the pulse wave at these different depths can be resolved between channels.

In this study, the rPPG signals derived from the Red and Green/Blue channels were used to calculate MW-PTT features, denoted as PTT_{RG} and PTT_{RB}, respectively. The calculation pipeline involved the following steps:

- Band-pass filtering: A 4th-order Butterworth bandpass filter with a frequency range of [0.6, 3.0] Hz was applied to suppress low-frequency drift and high-frequency noise.
- Temporal upsampling: To enhance the temporal resolution of peak detection, the raw 100 FPS signals were temporally upsampled to 200 FPS using cubic spline interpolation.
- Peak detection: Systolic peaks were detected in both the phase-advanced and phase-lagged rPPG signals using MATLAB's **findpeaks** function. The MW-PTT was defined as the time difference between corresponding peaks in the same cardiac cycle, with the constraint that the phase-lagged peak must occur within 20 samples (100 ms) after the phase-advanced peak.
- Segment validation: For each 30 s sliding window, valid PTT estimates were retained if their count exceeded one-sixth of the total number of phase-lagged peaks. Segments failing to meet this criterion were discarded.

III. RESULTS AND DISCUSSION

In this section, the results and discussion of the spatial distribution of rPPG features are presented. The results of each metric are composed of two main parts: heatmaps are utilized to visualize representative spatial patterns, while boxplots quantify the statistical trends across 15 subjects. For the quantitative analysis, the ROI was segmented into 10 equidistant bins along the vertical axis. The x-axis of the boxplots represents the normalized distance from the light source, ranging from 10% (closest to the source) to 100% (furthest from the source).

A. Spatial Distribution of AC, DC and AC/DC

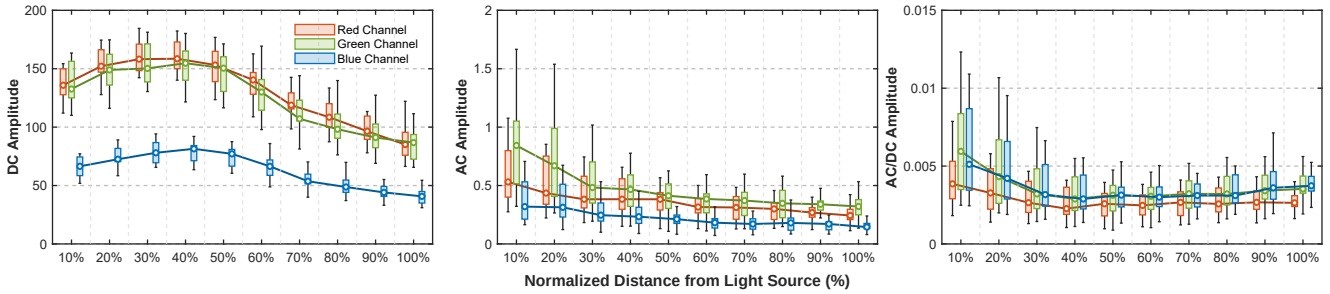
Fig. 3 shows the spatial trends relative to the normalized distance, while Fig. 4 exemplifies the representative spatial patterns.

1) Comparison between Illumination Modes: Under non-contact illumination, the overall spatial distribution of all metrics appears relatively uniform compared to the contact illumination condition. The DC component reaches its maximum at the central finger pad region and attenuates towards the periphery. Conversely, the AC component appears relatively stronger at the finger edges, indicating enhanced pulsatile contrast where DC intensity is lower, likely due to the thinner tissue at the periphery, which allows for higher light transmission. Consequently, the AC/DC presents a trend inverse to the DC distribution, showing higher amplitudes at the edges.

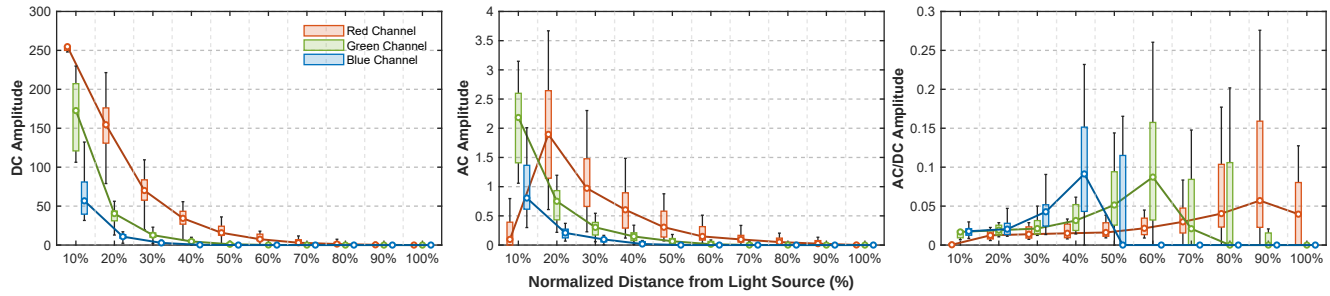
In contrast, the contact mode exhibits a strong dependence on the distance from the light source. Both DC and AC generally exhibit a monotonic decay as the distance from the source increases, as consistently observed in both heatmaps and boxplots. However, within the proximal region (0–10% normalized distance), the AC component exhibits an initial increase followed by a decrease. The near-zero AC amplitude at the very beginning is primarily caused by pixel saturation, which suppresses pulsatile signal variations. As the influence of overexposure diminishes with increasing distance, the AC amplitude rises before subsequently declining due to attenuation. This overexposure region is wavelength-dependent, being most pronounced in the Red channel and least evident in the Blue channel. Notably, although AC decreases with distance, the DC component decays at a relatively faster rate, exhibiting a characteristic rise–then–fall pattern across all channels of the AC/DC component.

Beyond the spatial patterns, an apparent difference lies in the signal magnitude. Contact illumination yields higher signal amplitudes compared to non-contact mode. This disparity is most pronounced in AC/DC, which is significantly ($10\times$) higher in contact mode than that in non-contact mode. This contrast arises because non-contact imaging integrates photons originating from a broad range of tissue depths, losing the "banana-shaped" propagation path present in contact mode. As a result, signals from superficial layers lacking pulsatile components are mixed with those from deeper arterial layers, thereby diluting the measured pulsatile amplitude and reducing sensitivity. In addition, surface specular reflections further elevate the DC baseline without contributing to pulsatile information, making it more challenging to extract the physiological signal compared to the contact modality.

2) Comparison between Wavelength Channels: The spatial trends in non-contact mode remain consistent across the Red, Green, and Blue channels, differing mainly in absolute amplitude, which can be attributed to the wavelength-dependent optical properties of skin. This suggests that wavelength-induced spatial variations are secondary under non-contact conditions, where illumination geometry is relatively uniform. The DC intensity follows the order of $R > G > B$. This occurs because Red light falls within the optical window where absorption by melanin and hemoglobin is minimal, leading to high backscattering, whereas Green and Blue light are strongly attenuated by epidermal pigments and blood [16]. Conversely, the AC amplitude is highest in the Green channel ($G > R > B$). This aligns with the fact that hemoglobin exhibits strong absorption peaks in the green spectral range ($500 - 600$ nm), maximizing the pulsatile contrast [2]. Although Blue light is also strongly absorbed by blood, its shallow penetration depth limits its interaction with the pulsatile arterioles. Regarding the AC/DC, the Green and Blue channels exhibit higher values than the Red channel

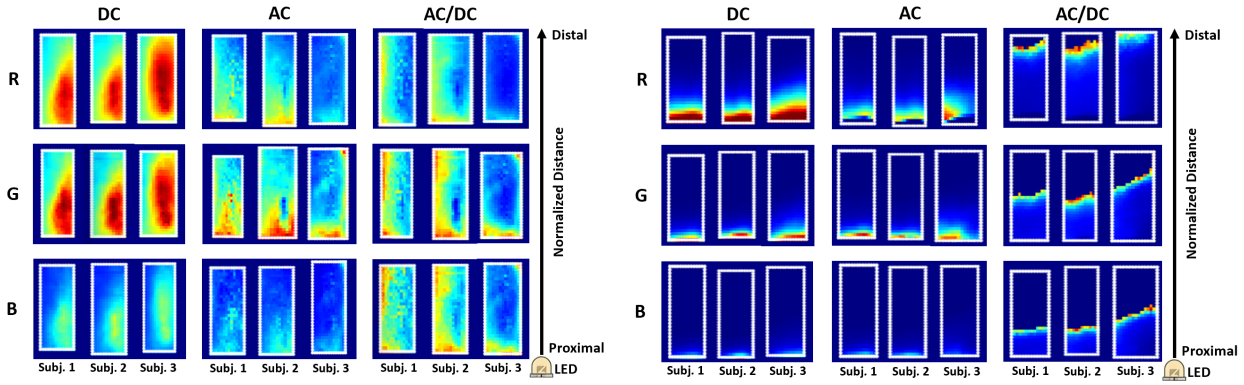


(a) Spatial Distribution of AC, DC and AC/DC under Non-Contact Illumination



(b) Spatial Distribution of AC, DC and AC/DC under Contact Illumination

Fig. 3: Statistical analysis of signal metrics across 15 subjects. The boxplots quantify the spatial distribution of DC, AC, and AC/DC amplitudes relative to the normalized distance from the light source (x-axis: 10% represents the region closest to the light source/mask bottom, 100% represents the furthest).



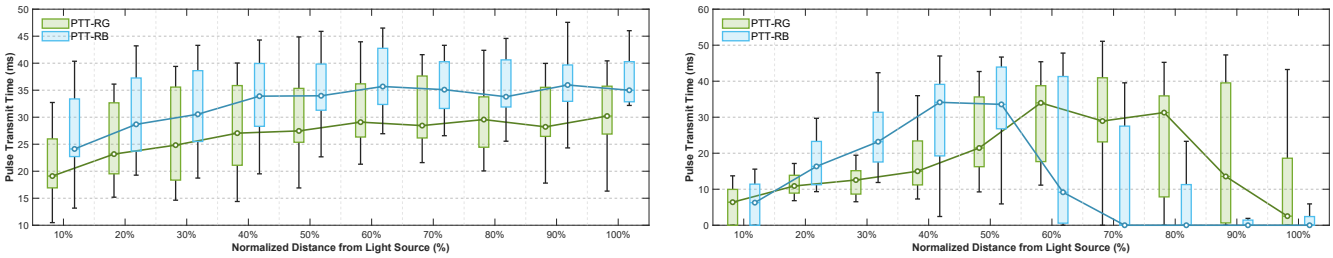
(a) Non-Contact Illumination

(b) Contact Illumination

Fig. 4: Representative spatial heatmaps illustrating the distribution of signal metrics. (a) Non-Contact mode and (b) Contact mode. Each white patch corresponds to the finger ROI for an individual subject. The light source is located at the bottom-center for contact mode and bottom-right for non-contact mode of the ROI. Color limits are consistent across channels for direct comparison. The subject layout is identical across all subfigures (i.e. three subjects are exemplified).

($G \approx B > R$). For the Red channel, the high DC baseline acts as a large denominator, suppressing the normalized ratio. In contrast, the Blue channel's extremely low DC baseline mathematically compensates for its weak pulsatile signal.

The contact mode displays significant wavelength-dependent behaviors attributed to optical penetration depth, with its signal attenuation rate following the order of $R < G < B$. This divergence is most evident in the AC/DC metric. Specifically, the AC/DC peaks occur at approximately 90% normalized distance for the Red channel, 60% for the Green channel, and 40% for the Blue channel, reflecting progressively shorter effective propagation distances for shorter wavelengths. This behavior is hypothesized to be related to wavelength-dependent penetration depth. Shorter wavelengths possess significantly higher absorption coefficients in blood and melanin compared to the Red light [17]. Consequently, photons at shorter wavelengths penetrate shallower, exhibiting shorter effective propagation distances. Once the propagation limit is exceeded, the AC/DC signal diminishes toward zero, giving rise to the observed rise-and-fall trend.



(a) Spatial Distribution of MW-PTT under Non-Contact Illumination

(b) Spatial Distribution of MW-PTT under Contact Illumination

Fig. 5: Statistical analysis of Multi-Wavelength Pulse Transit Time (MW-PTT) across 15 subjects. The boxplots quantify the spatial distribution of PTT_{RG} and PTT_{RB} relative to the normalized distance from the light source (x-axis: 10% represents the region closest to the light source, 100% represents the furthest).

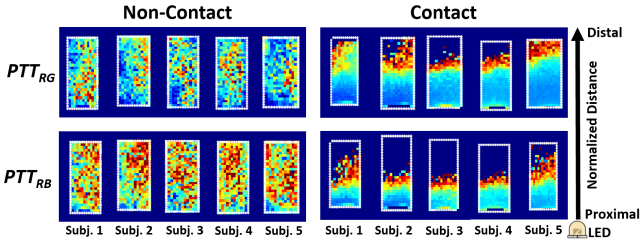


Fig. 6: Representative spatial heatmaps illustrating the distribution of PTT_{RG} and PTT_{RB} under contact and non-contact illumination modes. Each white patch corresponds to the finger ROI for an individual subject. Light source is located at the bottom-center for contact mode and bottom-right for non-contact mode of the ROI. The subject layout is identical across all subfigures (i.e. five subjects are exemplified).

B. Spatial Analysis of MW-PTT

The PTT_{RG} and PTT_{RB} were computed to evaluate the spatial distribution. To provide a statistical perspective, Fig. 5 quantifies the MW-PTT variations relative to the normalized distance from the light source, while Fig. 6 visualizes the representative spatial patterns.

Under non-contact illumination, MW-PTT exhibits a consistent spatial trend across wavelength pairs, characterized by a slight increase at short normalized distances, followed by a stable stage. This initial increase is possibly attributed to the reduced tissue thickness and increased surface curvature near the fingertip, which shortens the effective optical path length. As the distance increases, tissue thickness becomes more uniform and surface curvature flattens, resulting in stabilized MW-PTT. Across channel pairs, the overall spatial trends remain highly consistent. The phenomenon that the magnitude of PTT_{RB} consistently exceeds that of PTT_{RG} can be explained by the differential optical penetration depths. MW-PTT essentially measures the phase delay of the pulse wave arriving at different vascular depths. Red light penetrates into the deepest arterioles, blue light interacts with the most superficial capillaries, and green light reaches an intermediate vascular depth. Accordingly, the depth difference between the Red and Blue channels is physically larger than that between the Red and Green channels. Therefore, a larger phase delay is observed between the Red and Blue channels.

In contrast, MW-PTT under contact illumination demonstrates a pronounced distance-dependent behavior. With increasing distance from the light source, the extended optical propagation path delays pulsatile signal arrival, producing an initial increase in MW-PTT. Beyond the wavelength-dependent propagation limit, MW-PTT decreases toward zero as the pulsatile signal becomes insufficient for reliable phase estimation, resulting in a clear rise-and-fall spatial pattern. Notably, the peak positions of MW-PTT differ between wavelength pairs. PTT_{RB} reaches its maximum at approximately 40% normalized distance, whereas PTT_{RG} peaks at around 60%. Importantly, these peak locations closely align with the corresponding AC/DC maxima of the Blue and Green channels, respectively. This further supports that MW-PTT spatial distribution under contact illumination is primarily governed by wavelength-dependent optical penetration depth. Specifically, Green light exhibits lower absorption by superficial pigments (e.g., melanin) and skin tissue compared to Blue light, resulting in a greater penetration depth and extended propagation distance [17].

C. Discussion and Envision

The primary contribution of this work is the visualization of rPPG signal formation through high-resolution spatial heatmaps, providing direct evidence on how illumination geometry shapes physiological signals. This visual approach allows us to map the transition from contact mode governed by diffusive transport to non-contact mode dominated by superficial backscattering. The contrasting spatial distributions observed between the two illumination modalities underscore the fundamental role of the light source to rPPG measurement. While non-contact mode yields spatially uniform signal quality independent of wavelength, contact mode creates a distance-dependent pattern. The observed rise-and-fall pattern in AC/DC and MW-PTT confirms that signal quality is a trade-off between penetration depth and signal intensity. At the proximal source, the detector is blinded by non-pulsatile saturation, while the signal is flooded by noise at distal regions. The optimal ROI selection lies where photons have penetrated sufficiently deep to reach the arterioles but

still retain sufficient intensity for reliable detection.

Furthermore, the spatial alignment between the peaks of MW-PTT and AC/DC under contact illumination confirms the physical limits imposed by wavelength-dependent optical penetration. Due to higher absorption coefficients, Blue and Green light cannot reach distal ROIs [17]. This implies that contact illumination imaging can possibly provide depth-sensitive information, where radial distance from the source acts as an indicator for penetration depth.

In summary, this study highlights the distinct trade-offs inherent to each illumination modality. Non-contact mode offers spatial uniformity and robustness, as its uniform signal distribution makes it resilient to ROI misalignment and tracking errors. However, this convenience comes at the cost of low modulation depth due to the dominance of shallow-layer non-pulsatile reflections and surface specular reflections. Conversely, contact-based illumination yields signals with significantly higher amplitudes and depth-resolved information, yet it demands precise ROI selection to avoid saturation or attenuation artifacts corresponding to the precise source–detector separation control on contact PPG devices. These insights suggest that, based on the understanding of wavelength-dependent source–detector separation mechanisms in tissue, the quality of rPPG imaging for unobtrusive monitoring can be optimized through control of illumination geometry, such as using speckle-dot structured light patterns that mimic contact-based point sources to improve the signal quality while still retaining the merits of non-contact sensing. Furthermore, our findings regarding wavelength-specific peak shifts provide a physical basis for dynamic ROI selection, allowing future studies to optimize SpO_2 and blood pressure estimation accuracy by adapting different sampling windows to target distal regions for red light and proximal for blue/green channels.

IV. CONCLUSIONS

This study utilized a unified camera-based framework to visualize the spatial characteristics of rPPG signals under contact and non-contact illumination. Through spatial heatmap analysis, we demonstrated that illumination geometry fundamentally determines the spatial distribution and quality of hemodynamic signals. Non-contact mode behaves as a planar wave, producing uniform spatial distributions of AC, DC, AC/DC and MW-PTT. Contact mode acts as a point light source, introducing strong distance- and wavelength-dependent patterns. Notably, under contact illumination, the normalized AC/DC and MW-PTT exhibit similar rise-and-fall trends, with peak locations closely align across wavelengths, indicating shared physical constraints related to wavelength-dependent penetration depth and effective propagation distance. Specifically, the maximum amplitudes occur at 90%, 60%, and 40% of the normalized fingertip distance for Red, Green, and Blue channels, respectively. These results suggest that optimal signal extraction requires targeting specific regions where photon path length balances penetration depth against signal attenuation. Future work will investigate the effects of dynamic physiological perturbations on Ratio-of-Ratios-based SpO_2 estimation and MW-PTT-based blood pressure assessment under both illumination modes. Furthermore, based on the spatial and optical insights gained from this study, we aim to explore advanced illumination geometries to optimize rPPG signal fidelity.

REFERENCES

[1] J. Allen, "Photoplethysmography and its application in clinical physiological measurement," *Physiological measurement*, vol. 28, no. 3, p. R1, 2007.

[2] W. Verkruysse, L. O. Svaasand, and J. S. Nelson, "Remote plethysmographic imaging using ambient light." *Optics express*, vol. 16, no. 26, pp. 21 434–21 445, 2008.

[3] Y. Ye, L. Pan, D. Yu, D. Gu, H. Lu, and W. Wang, "Notch rgb-camera based spo2 estimation: a clinical trial in neonatal intensive care unit," *Biomedical optics express*, vol. 15, no. 1, pp. 428–445, 2023.

[4] H. Shu, H. Lai, H. Liu, and W. Wang, "Camera-spo 2 measurement using time-division multiplexed near-infrared imaging: A feasibility study on dc-based calibration." in *Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE Engineering in Medicine and Biology Society. Annual International Conference*, vol. 2025, 2025, pp. 1–7.

[5] Y. Huang, D. Huang, J. Huang, G. Wang, L. Pan, H. Lu, M. He, and W. Wang, "Camera-based blood pressure monitoring based on multi-site and multi-wavelength pulse transit time features," *IEEE Transactions on Instrumentation and Measurement*, 2024.

[6] Y. Zeng, Y. Zhu, X. Song, Q. Wang, J. Yang, and W. Wang, "Camera-based neonatal blood pressure estimation from multisite and multiwavelength pulse transit time-a proof of concept in nicu," *IEEE Internet of Things Journal*, 2025.

[7] J. Liu, Y.-T. Zhang, X.-R. Ding, W.-X. Dai, and N. Zhao, "A preliminary study on multi-wavelength ppg based pulse transit time detection for cuffless blood pressure measurement," in *2016 38th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. IEEE, 2016, pp. 615–618.

[8] V. V. Tuchin *et al.*, "Tissue optics." Society of Photo-Optical Instrumentation Engineers (SPIE) Bellingham, WA, USA, 2015.

[9] F. Martelli, T. Binzoni, S. Del Bianco, A. Liemert, and A. Kienle, "Light propagation through biological tissue and other diffusive media: theory, solutions, and validations." SPIE Bellingham, WA, USA, 2022.

[10] G. De Haan and V. Jeanne, "Robust pulse rate from chrominance-based rppg," *IEEE transactions on biomedical engineering*, vol. 60, no. 10, pp. 2878–2886, 2013.

[11] W. Wang, A. C. Den Brinker, S. Stuijk, and G. De Haan, "Algorithmic principles of remote ppg," *IEEE Transactions on Biomedical Engineering*, vol. 64, no. 7, pp. 1479–1491, 2016.

[12] H. Shao, L. Luo, J. Qian, M. Yan, S. Chen, and J. Yang, "Remote photoplethysmography in real-world and extreme lighting scenarios," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 10 858–10 867.

[13] S. Yi, D. Yu, Y. Zhu, and W. Wang, "Illumination-robust camera-ppg via skin-guided auto-exposure," in *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. IEEE, 2024, pp. 1–4.

[14] K. Budidha, S. Chatterjee, M. Qassem, and P. A. Kyriacou, "Monte carlo characterization of short-wave infrared optical wavelengths for biosensing applications," in *2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*. IEEE, 2021, pp. 4285–4288.

[15] L. Finlayson, I. R. Barnard, L. McMillan, S. H. Ibbotson, C. T. A. Brown, E. Eadie, and K. Wood, "Depth penetration of light into skin as a function of wavelength from 200 to 1000 nm," *Photochemistry and Photobiology*, vol. 98, no. 4, pp. 974–981, 2022.

[16] R. R. Anderson and J. A. Parrish, "The optics of human skin," *Journal of investigative dermatology*, vol. 77, no. 1, pp. 13–19, 1981.

[17] A. N. Bashkatov, E. A. Genina, V. I. Kochubey, and V. Tuchin, "Optical properties of human skin, subcutaneous and mucous tissues in the wavelength range from 400 to 2000 nm," *Journal of Physics D: Applied Physics*, vol. 38, no. 15, p. 2543, 2005.